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January 9, 1975
SUPERSEDING
Fed. Spec. GGG-D-631B
January 30, 1961

FEDERAL SPECIFICATION

DRESSER, ABRASIVE WHEEL, HAND, REVOLVING CUTTER TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated abrasive wheel dressers of the star-shaped revolving cutter type.

1.1.1 Federal specification coverage. This specification does not cover all varieties of dressers that are commercially available but is intended to include only those varieties generally purchased by the Federal Government.

1.2 Classification.

1.2.1 Type and sizes. Dressers shall be of one type and of the three sizes shown in table I (see 6.2). The cutter sets shall be of one type and of the three sizes shown in table II (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

PSC 5120

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of dressers otherwise conforming to this specification.

3.2 Material. The chemical composition of the materials and the heat treatment employed shall be such as to produce dressers complying with the requirements specified herein. All forgings shall be of uniform quality, free from all defects and imperfections which may affect their serviceability and durability.

3.3 Design. The dresser shall be similar to figure 1 and conform to the applicable data shown in table I for the size specified (see 6.2). The dresser shall consist essentially of a dresser body, cutter set, and bearing assembly, and shall be constructed to provide a firm mounting for revolving cutter sets, and permit simple replacement of the cutter set.

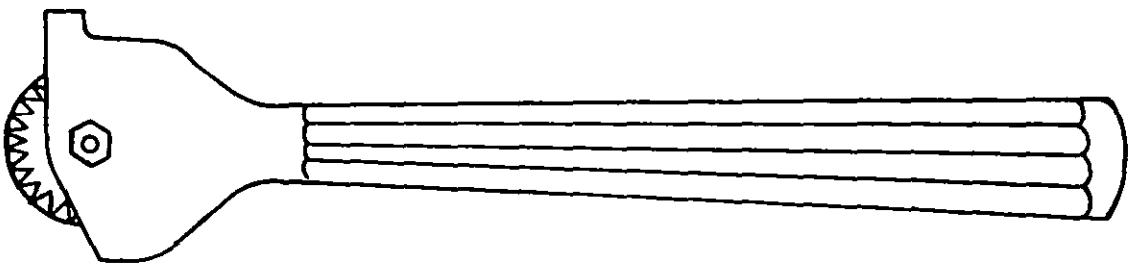


FIGURE 1. Dresser.

TABLE I. Dresser

Size	Outside diameter of cutter set	Weight	Overall length	Recommended for use on grinding wheels with	
	Nominal	Maximum	Nominal	Diameters	Face widths
	Inches	Ounces	Inches	Inches	Inches
0	1-1/4	30	11	Up to 10	Up to 2
1	1-1/2	35	12	10 to 16	2 to 3
2	2-3/8	35	13	Over 16	Over 3

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3.3.1 Dresser body. The body shall be made of cast or drop-forged steel, fine-grained malleable iron, cast iron, or an alloy, and shall be forged or cast as a single piece. The body shall consist of a suitable handle affording a comfortable grip, which shall be connected by a substantial shank to the cutter head with its protective hood. The hood or dresser head shall be so designed that when used, shall deflect the abrasive dust away from the operator. The hood or dresser head shall be constructed to permit full use of cutter teeth to their root diameter. Suitable means of support shall be provided on the underside of the head for bearing on the tool rest. The body shall be free from fins, burrs, excessive roughness or sharp corners, and shall have any one of the following finishes:

(a) Natural, free from excess oxide scale, oil treated.

(b) Chemically produced oxide coating, oil treated.

(c) Chemically produced phosphate coating, oil treated.

(d) Zinc plated.

(e) Lacquered or enameled.

3.3.2 Cutter set. The cutter set shall conform to the applicable data shown in table II and replacement sets shall be of the size specified (see 6.2). Each set shall have two cutters placed face to face in the center, flanked on each side by a separator. The additional cutters, as required for the respective set, shall be distributed equally on each side of the set. Cutters shall be of the star-tooth design and shall be made of high-carbon tool steel having a hardness of Rockwell C 55 to 60. The bore of the cutters and separators shall be concentric with the outside diameter within 0.020 inch. Separators shall be of high-grade sheet steel.

TABLE II. Cutter sets

Size	Outside diameter	Cutters per set	Separators per set	Bore plus 1/64 minus 0	Thickness of cutter	Number of teeth	Total depth of teeth	
					Minimum		Minimum	Maximum
	Inches			Inch	Inch		Inch	Inch
0	1-1/4 (±1/16)	4	2	1/4	0.085	16 to 18	1/4	5/16
1	1-1/2 (±1/8)	4	2	1/2	.085	16 to 18	5/16	3/8
2	2-3/8 (±1/8)	6	2	19/32	.115	16 to 18	7/16	1/2

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3.3.3 Bearing assembly. The bearing assembly shall consist essentially of a spindle and bushings or a multiple hole bearing block securely retained within the dresser head by means of externally threaded bushing retainers or steel plate side retainers. If threaded bushing retainers are used, the retainer having a left-hand thread shall be clearly marked to so indicate with a letter "L". If steel plate side retainers are used, they shall be securely attached to the dresser head by machine screws or bolts. The spindle shall be either smoothly finished its full length or threaded on one end suitable for assembly in the dresser head and to accept the cutter set or bushing. These components shall be heat-treated to a hardness of Rockwell C 55 to 60. Where necessary, for the proper operation of the dresser, spacers shall be provided.

3.4 Marking. Each dresser shall be marked in a plain and permanent manner with the manufacturer's name or a trademark of such known character that the source of manufacture may be readily determined.

3.5 Workmanship. The dresser shall be free from any defects which may affect serviceability, durability, or life expectancy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling for examination and test.

4.2.1 Inspection lot. An inspection lot shall consist of all dressers or cutter sets of one size offered for acceptance at one time. The sample unit shall be one dresser or one cutter set.

4.2.2 Sampling for examination and test. Sampling for the examination and test shall be accomplished in accordance with MIL-STD-105 at inspection level II with an Acceptable Quality Level (AQL) of 4.0.

4.3 Examinations.

4.3.1 Each sample selected in accordance with 4.2.2 shall be examined for conformance to the requirements for material, design, weight, dimensions, finish, markings, and workmanship. Any sample that does not comply with all requirements shall be classified as a reject.

4.3.2 Examination of preparation for delivery. Examination shall be made to determine compliance with preservation, packaging, packing, and marking requirements in accordance with PPP-P-40.

4.4 Test.

4.4.1 Hardness test. Each sample shall be tested for the specified hardness. Test impressions on the cutters shall be taken approximately 1/4 inch from the cutting edge and the impressions on the bearing assembly may be taken at any desired location.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The dressers shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. Preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.2), and the marking shall be civil or military agency as specified (see 6.2).

6. NOTES

6.1 Intended use. Dressers covered by this specification are intended to be used to reface grinding wheels by removing the worn portion of the wheel in accordance with the instructions in ANSI B7.1.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size number of dresser required (see table I).
(c) Size number of cutter set required (see table II).
(d) Level of preservation, packaging and packing required (see 5.1).

MILITARY INTEREST:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity

GSA-FSS

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Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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